Interpreting benthic Common Era foraminiferal records under the assumption of modern-day circulation

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Key Points:

- We present an inversion method that incorporates age model uncertainty to infer far-field properties from benthic sediment core records (137/140 characters)
- The reconstructed North Atlantic Little Ice Age is characterized by a 0.11-1.2K/kyr cooling (1σ confidence interval) from 1200-1900CE (140/140 characters)
- A long-term cooling in the Nordic Seas is reconstructed, and is found to be approximately two times greater than that of the subpolar gyre (138/140 characters)

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Abstract

Paleoceanographic reconstructions of the past 2,000 years indicate a large-scale cooling in the latter half of the last millennium, but significant conflict on magnitude and timing exist. This conflict is partially due to the inference of surface conditions from sparse, noisy, near-surface marine proxies that can exhibit seasonal biases. Here, we develop a novel methodology for inferring far-field changes from benthic sediment core records via an ocean circulation model. Benthic sediment core records are not subject to high-energy near-surface variability or seasonal biases and record much more stable signals than equivalent planktonic records. We present a possible surface temperature and $\delta^{18}O_{seawater}$ history that fits a dataset of 11 high-resolution benthic $\delta^{18}O_{calcite}$ records. The estimate presented here is within error of instrumental records, and is characterized by a Little Ice Age sea surface temperature cooling of 0.67 ± 0.56 K/kyr, which is similar to, but at the high end, of previous independent estimates obtained from Common Era marine surface records. Our reconstruction is characterized by a larger magnitude millenial-scale cooling in the Nordic Seas (0.9 ± 2.7 K) than in the subpolar gyre (0.5 ± 2.2K), indicative of an Arctic amplification of LIA cooling.

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1 Introduction

The past 2,000 years, referred to as the Common Era (CE), is a critical period for understanding the response of the Earth's climate to perturbations on long timescales. Climatic perturbations during the CE include decreasing solar irradiance due to the precession of the equinoxes and stratospheric sulfate injection from volcanic eruptions. Paleoclimatic and paleoceanographic reconstructions of the CE indicate that large-scale climate deviations were comparably small when compared to changes that occurred on longer timescales, such as the deglaciation. However, notable regional- to global-scale climate events have been identified in various proxy records. These include the Little Ice Age (LIA), a period of anomalously cool temperatures relative to the CE mean, or the Medieval Warm Period (MWP), a period of anomalously warm temperatures that may have preceded the LIA. Understanding the spatiotemporal extent of these climate events is a critical step in constraining the response of the climate system to various perturbations on timescales longer than that of the instrumental record.

Reconstructions of sea surface temperature (SST) have indicated a global decrease from 1-1800 CE, which has been attributed to a thermodynamic response to high-frequency volcanism (McGregor et al., 2015). Other oceanic or atmospheric feedback mechanisms that may have reinforced cooling have been suggested, such as an increase of Arctic Ocean sea-ice as a direct response to decadally-paced volcanic eruptions beginning in 1257-1258CE, which ultimately leads to a freshening of the subpolar North Atlantic that weakens convection (Zhong et al., 2010; Miller et al., 2012; Lehner et al., 2013; Schleussner & Feulner, 2013). Weakened convection has been associated with a slowdown of the Atlantic Meridional Overturning Circulation (AMOC), although some modelling studies have suggested that an AMOC slowdown is not necessary to explain the full magnitude of LIA cooling (Moreno-Chamarro et al., 2016). Other suggested reinforcement mechanisms include large-scale shifts in the North Atlantic Oscillation (NAO) as part of a feedback system related to a weakened Atlantic Meridional Overturning Circulation (AMOC) during the LIA (Trouet et al., 2009), although the robustness of NAO reconstructions has been questioned (Ortega et al., 2015).

Paleoclimate proxy records and reconstructions have been used to investigate the validity of many of these ideas. However, significant confusion still exists on overturning reconstructions over the CE. Notably, through the usage of sortable silt records, no significant centennial-scale changes prior to 1800 have been detected in the deep western boundary currents associated with the Labrador Sea (Thornalley et al., 2018) or the Nordic Seas (Moffa-Sanchez et al., 2015), both of which are important components of the lower limb of the AMOC. These findings oppose that of Lund et al. (2006), where geostrophic reconstructions of the Florida Current indicated weakening from 1200-1850 CE, suggesting a slowdown of the northward transport of tropical waters in the AMOC. In the more recent past, evidence for an AMOC slowdown over the 20th century has been found in Labrador Sea sortable silt records (Thornalley et al., 2018), as well as SST-fingerprint AMOC reconstructions (Rahmstorf et al., 2015). However, synchronous LIA and industrial changes in benthic and planktonic oxygen isotope records from ∼1300m-2200m indicate the presence of a largely unchanging AMOC over the CE (Lu et al., 2023).

The mechanism outlined by Zhong et al. (2010) of a sea-ice expansion in the Arctic has been supported by a ∼1400 CE shift towards Arctic-origin ice-rafted debris in the Labrador Sea (Alonso-Garcia et al., 2017). Significant LIA changes in subpolar gyre properties have been documented, although the precise timing is unclear. Destabilization of the subpolar gyre has been suggested to occur ∼1250 CE by deep-sea coral records (Copard et al., 2012) and between 1330-1380 by bivalve resilience records (Arellano-Nava et al., 2022). A 1400-1450CE LIA weakening of the subpolar gyre has also been inferred from planktonic foraminiferal oxygen isotope reconstructions (Moffa-Sánchez, Born, et al., 2014; Moffa-Sanchez & Hall, 2017). However, foraminiferal faunal assemblages, abundances, and isotopic trends from the Reykjanes Ridge indicate largely constant Holocene conditions, with a large-scale shift in subpolar gyre conditions beginning in the 1750s, but with peak changes occurring in the 20th century (Spooner et al., 2020).

Overall, studies do not show support for large-scale overturning changes during the LIA. However, there is more consistent support for changes in subpolar gyre strength, although the exact timing and magnitude is unclear. One of the primary limitations of our understanding of LIA dynamics comes from the point-wise nature of paleoceanographic proxy records. Inferring behavior of a large-scale oceanographic feature such as the subpolar gyre typically requires spatially dense and high temporal resolution measurements, which can be difficult to attain for paleo-timescales. Although there have been several different reconstructions of large-scale surface temperature variability over the CE (Esper et al., 2024), there have been very few purely marine proxy-based reconstructions of the CE. Early SST reconstructions such as that of Cunningham et al. (2013) use composite-based methods that rely upon modern-day climatologies that may not be indicative of CE dynamics. The Last Millenium Reanalysis (Hakim et al., 2016) reconstructs SST, but exhibits much less cooling than purely marine-based reconstructions such as Ocean2k (McGregor et al., 2015) (see Figure 7). These reconstructions all rely upon surface records, which experience a high degree of noise and seasonal effects. When trying to extract long-term climate signals, benthic records may be a better option, as they are not subject to high-frequency surface variability. Given that North Atlantic ocean dynamics are thought to be a primary component of positive feedback mechanisms reinforcing the LIA, obtaining a more spatially-complete estimate of near-surface variability is critical.

In this paper, we develop an inverse approach to inferring far-field properties from paleoceanographic proxy data through the usage of a data-informed circulation model. Here, we apply a modern-day circulation model under the assumption that there were no large-scale circulation changes over the CE. Although there is no clear answer on whether or not there were large-scale circulation changes over the CE, testing whether or not simultaneous sediment core records can be well-fit under the assumption of steady-state modern-day circulation has been previously used as a test for the necessity of circulation change (Burke et al., 2011; Gebbie, 2012; Amrhein et al., 2015).

The circulation model used here is a transient inversion of modern-day tracer observations (Gebbie & Huybers, 2012) previously used to constrain CE deep Pacific and Atlantic cooling (Gebbie & Huybers, 2019). This prior application estimated a temperature history that could explain the observed warming from the 1870s HMS Challenger expedition to modern hydrography. It has also been used to explain the lag between Atlantic and Pacific warming in deglacial δ^{18}O records (Gebbie, 2012). Here, we apply this model to CE benthic sediment core δ^{18}O records for the first time, and develop a methodology to maximize the spatial field reconstructed while minimizing propagated uncertainties. This is accomplished by decomposing surface source water maps for each core site into orthogonal spatial modes, and solving for the mode magnitude of surface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ through time that best explains the benthic $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ records. We separate the effects of temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ by using experimentally defined ratios (Marchitto et al., 2014). Our problem has no independent constraints on temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, so we enforce a linear relationship of 10°C/‰ based upon isotope-enabled climate model output.

We will first show the ability of this methodology to fit a dataset of 11 high-resolution benthic records of $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ from along the Reykjanes Ridge in Iceland. Successfully fitting this data implies that possible surface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ histories exist that can explain the sediment core data without necessitating changes in circulation. Next, we will compare our reconstruction to instrumental records of Labrador Sea Water temperature and salinity, and demonstrate that we reconstruct similar magnitudes of temperature and salinity variability over the past 100 years. Our reconstruction can also be compared to other paleoceanographic estimates, and we show that the estimated LIA North Atlantic cooling trend is comparable to the global SST trend of 0.54-0.38°C/kyr identified in the Ocean2k dataset. Finally, we will discuss the spatiotemporal evolution of the LIA in the North Atlantic, and demonstrate that the LIA can be represented as a long-term cooling trend with regional magnitudes and timing. In our solution, the Nordic Seas are shown to be the first region to begin cooling, at ∼1200CE, followed by the subpolar gyre in ∼1400CE. The timing of the subpolar gyre cooling is similar to other independent records (Moffa-Sánchez, Hall, et al., 2014). The Nordic Seas experiences the largest cooling magnitude (0.8 ± 2.7 K) and the subpolar gyre experiences a lesser magnitude cooling of (0.37 ± 2.1K). For both regions, the main characteristic is the long-term cooling, but they also exhibit shorter-timescale antiphased variability. The spatially and temporally distinct temperature minima observed here may explain the multiple temperature minima present in Greenland borehole temperature reconstructions (Dahl-Jensen et al., 1998). In addition, pre-industrial antiphasing is noted between the Nordic Seas and subpolar North Atlantic, similarly to that identified in (Miettinen et al., 2012).

2 Data

2.1 Sediment Core Records from Lu et al. (2023)

The inversion presented here is applied to a set of 11 high-resolution benthic foraminiferal $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ sediment core records (Figure 3) previously presented in Lu et al. (2023). These records were collected from along the Reykjanes Ridge south of Iceland, and form a depth transect from 1100-2300m. Previously these records were used to demonstrate that the deep ocean LIA transition, as indicated by the benthic records, occurred at 1350 ± 50 CE, while the surface ocean LIA transition, as indicated by planktonic records, occurred from 1100-1700CE. In addition, a larger magnitude LIA was observed in the benthic records than in the planktonic records, indicate of Arctic amplification of LIA temperature changes, transferred to the sediment core sites by an active AMOC (Lu et al., 2023).

Here, we incorporate age model uncertainty into the inverse problem by applying the BACON-generated age models (Blaauw & Christen, 2011; McKay et al., 2021) from Lu et al. (2023). In order to translate the 1000 age model realizations into covariance

matrices with uncertainty in the $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ dimension, each age model realization was linearly interpolated at a common 10-year resolution. Then, for each timestep, the mean $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ was computed, for any interpolated timestep that had more than 100 realizations available. The covariance between two arbitrary timesteps t_1, t_2 can be computed according to Equation 1, where $\delta^{18}\mathrm{O}_i(t_1)$ refers to the interpolated $\delta^{18}\mathrm{O}$ value at t_1 for realization i and M is the total number of realizations with data at t_1 and t_2. Realizations will have different start and end times due to the age modeling process, so there are not necessarily 1000 realizations at every timestep.

[NOTE: This may be a good spot for a supplementary figure more completely explaining this process.]

$$cov(t_1, t_2) = \frac{\sum_i^M \left(\delta^{18}\mathrm{O}_i(t_1) - < \delta^{18}\mathrm{O}(t_1) >\right)\left(\delta^{18}\mathrm{O}_i(t_2) - < \delta^{18}\mathrm{O}(t_2) >\right)}{M} \tag{1}$$

Lu et al. (2023) used two benthic species of foraminifera to measure $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. The $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ measurements were obtained from Uvigerina peregrina (U. peregrina) tests for the three shallowest cores (MC28A, MC26A, and MC25A), and from Cibicidoides wuellerstorfi (C. wuellerstorfi) tests for the other eight sediment cores. While C. wuellerstorfi calcifies at equilibrium, U. peregrina is isotopically heavier by 0.47‰ (Marchitto et al., 2014).

$\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ is dependent upon calcification temperature and the $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, which is, in turn, controlled by temperature and salinity. Ultimately, increases in $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ are caused by decreasing temperature and/or salinification. In Equation 2, we show the experimental relationship for C. wuellerstorfi calculated by Marchitto et al. (2014), where $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ is in ‰PDB and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ is in ‰SMOW. This relationship can be applied to U. peregrina, as long as the 0.47‰ offset is removed.

$$\delta^{18}\mathrm{O}_{\mathrm{calcite}} = (-0.224 \pm 0.002)\frac{‰}{K}T + (3.53 \pm 0.02)‰ + (\delta^{18}\mathrm{O}_{\mathrm{seawater}} - 0.27‰) \tag{2}$$

In order to complete an inversion, the values reported from both C. wuellerstorfi and U. peregrina records must be put on a common scale. Effective modern-day $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ can be computed from the equilibrium value of $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ and temperature from the inversion of Gebbie and Huybers (2010). These values can then be compared to the core-top values of $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. We find that, after applying the experimental offset to the U. peregrina sediment cores, the measured $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ values are variably offset from the computed effective $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. This variable offset,

2.2 Total Matrix Intercomparison (TMI), steady-state and transient versions

In order to relate signals at the sediment core sites to far-field ocean regions, an ocean circulation model must be applied. In Gebbie and Huybers (2010), a box inversion technique is used to compute the pathways connecting interior ocean grid boxes to the surface. The global ocean is decomposed into a $4^\circ \times 4^\circ$ grid with 33 vertical levels, and the pathways that connect each interior grid box to the surface are computed by inverting six conservative tracers. This results in a matrix, $\mathbf{A}_{\mathrm{ss}}$, which contains the contribution of every surface grid cell to every interior cell, assuming that the ocean is at a steady-state. In Gebbie and Huybers (2012), this result is extended by incorporating radiocarbon observations in $\mathbf{A}_{\mathrm{t}}$, which contains the rate at which surface grid cells are transmitting water to each interior cell.

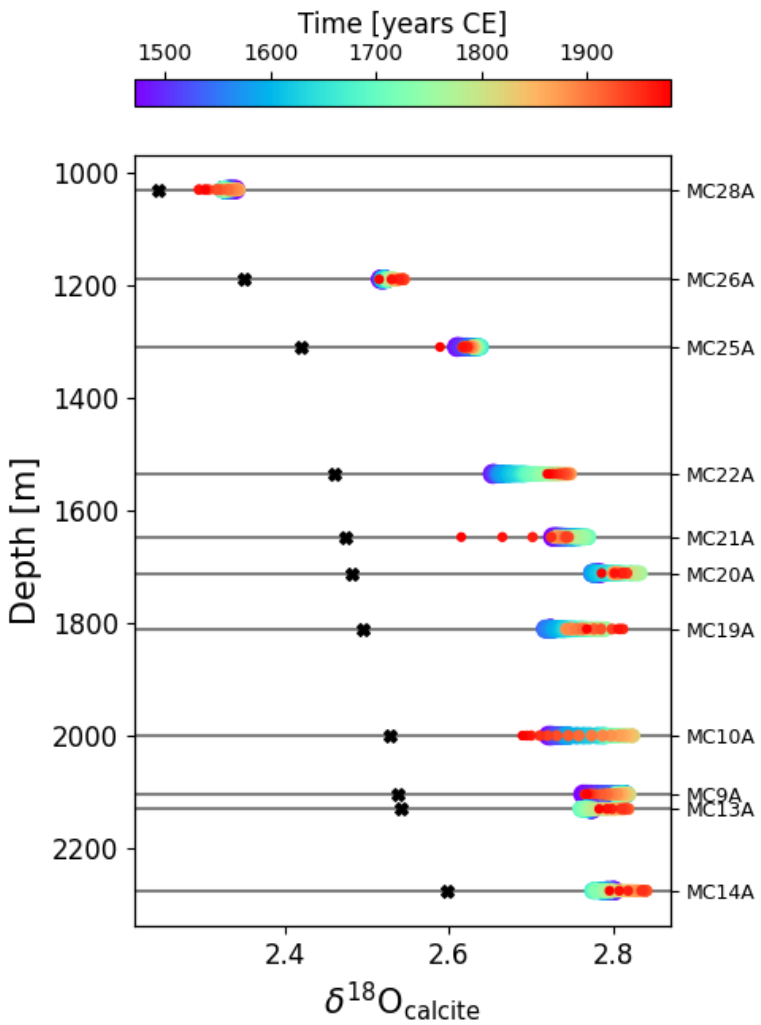

Figure 1. Rainbow dots, with size decreasing as time increases, show mean $\delta^{18}O_{calcite}$ for each sediment core, with 0.47‰ equilibration offset removed from U. peregrina records. Black crosses show effective $\delta^{18}O_{calcite}$ computed from the steady-state inversion of temperature and $\delta^{18}O_{seawater}$ from Gebbie and Huybers (2010) and observed at the sediment core sites. Computation of effective $\delta^{18}O_{calcite}$ is according to Equation 2.

2.3 Coupled Climate Model Output

SST fields from historical realization (1850-2015CE) 'r11i1p1f1' of the Community Earth System Model, version 2 (CESM2) (Danabasoglu et al., 2020) were used to constrain the relative variance of surface spatial modes.

The relationship between temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ was obtained using output from the iCESM1.2 coupled climate model (Brady et al., 2019). The 'TEMP' and 'R18O' variables from the large ensemble pre-industrial fully coupled control run were observed at the grid cell closest to the mean sediment core position, and the linear least squares relationship of the two timeseries was found to be 10°C/‰.

3 Methods

3.1 Overview

This inverse problem is characterized by inferring some time-varying far-field properties $\mathbf{u}$ from data at the sediment core sites $\mathbf{y}$. The connection between the sediment core sites and any region of the ocean can be computed by applying the transient TMI matrix $\mathbf{A}_{\mathrm{t}}$. Although this approach could be used to infer subsurface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, here we infer surface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, as it can more readily be compared to other records.

We are concerned with solving for the time-varying evolution of some collection of far-field patterns. These patterns could be oceanographic regions or features, and could be surface or subsurface. In order to reduce the dimensionality of the problem, we solve for the magnitude of surface spatial modes, obtained from a singular value decomposition of source water maps. The contribution of every surface grid cell to each sediment core site can be computed by computing $\mathbf{A^T}$ m These patterns can be propagated through the transient TMI matrix $\mathbf{A_t}$ and observed at the sediment core sites.

3.

$$\mathbf{y} = \mathbf{E}\mathbf{u} \tag{3}$$

3.2 Spatial Modes and Expected Variance

- How are spatial modes calculated?
- Expected variance of spatial modes in CESM
- Relationship between temperature and $\delta^{18}\mathrm{O}$ at core sites in iCESM

3.3 Computing Model Matrix

- Propagating modes through matrix
- Impulse response technique

3.4 Solving

- TWLS

3.5 Analysis techniques

- Surface reconstruction
- Reconstruction at core sites
- Cost functions

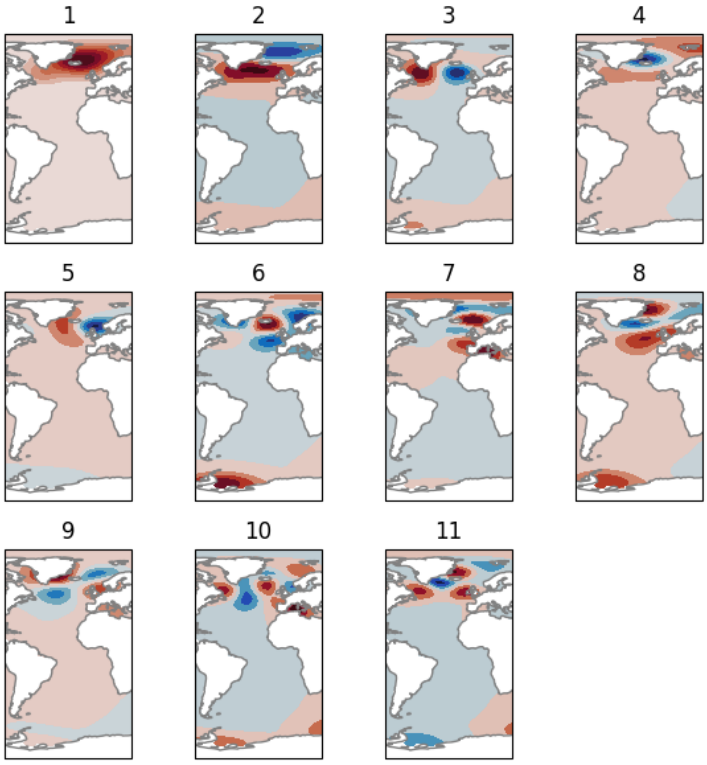

Figure 2. Surface spatial modes

4 Results

4.1 Data Fit

Statistics about how well the data is fit

4.2 Solution Comparison to Instrumental Record

4.3 Common Era Solution

5 Conclusion

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors

Acknowledgments
Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

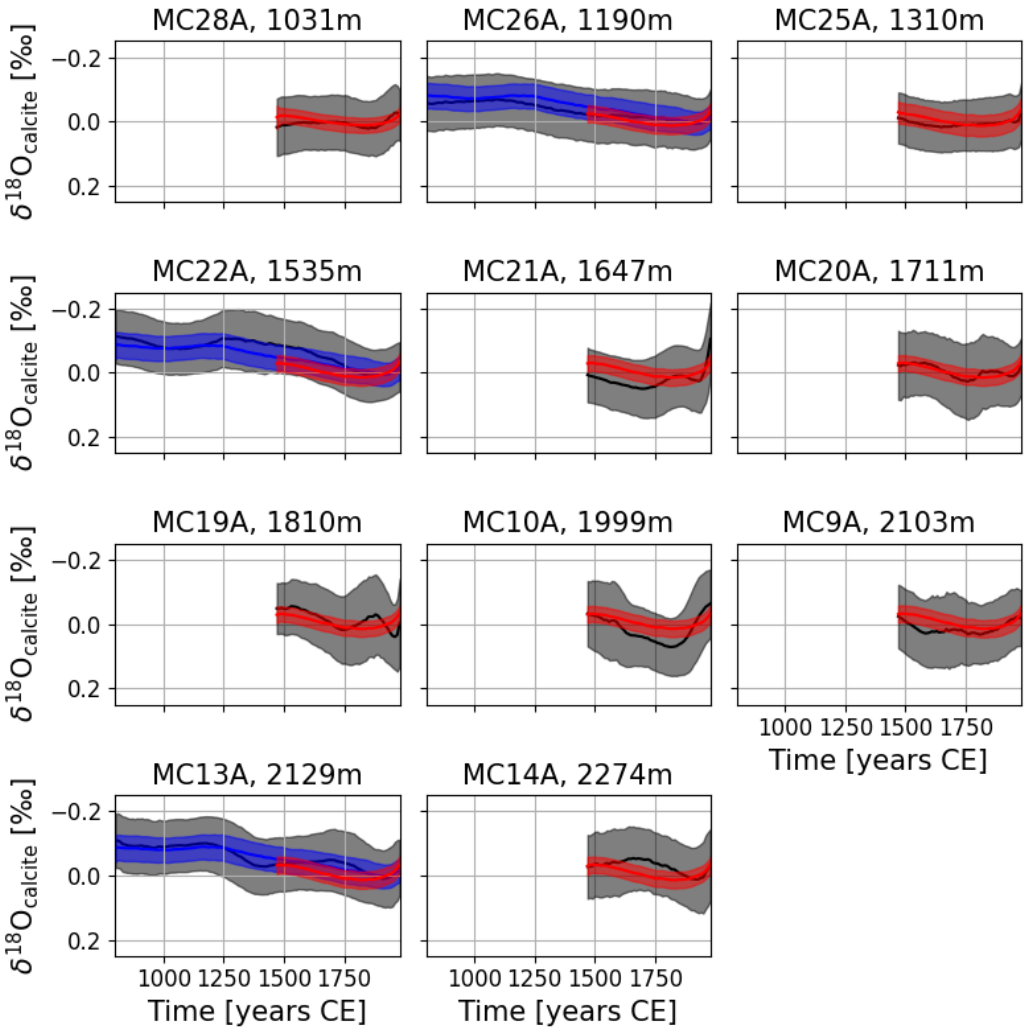

Figure 3. Reconstruction at core sites for all core estimate (red) and three oldest core estimate (blue). Black line shows sediment core record.

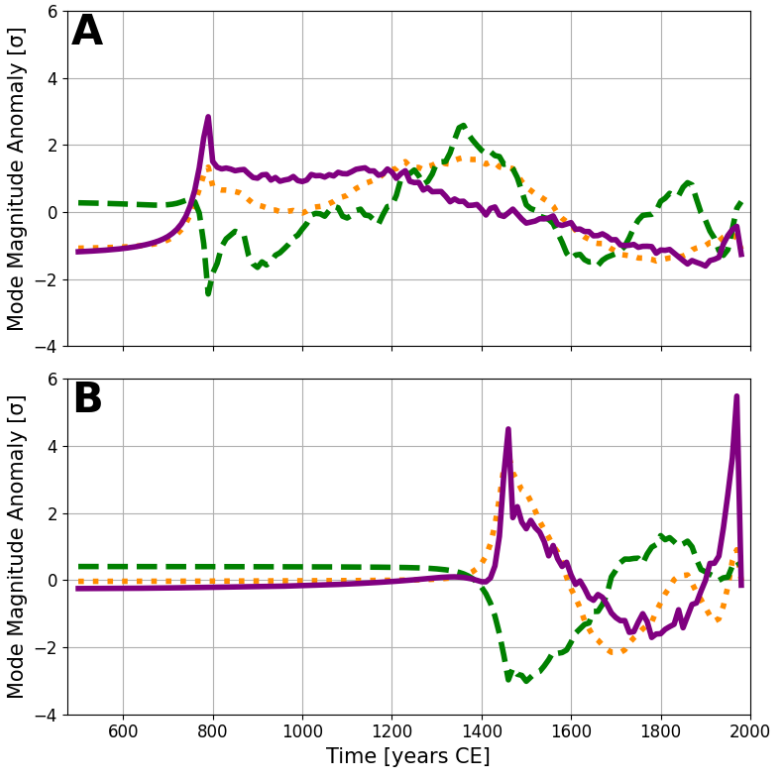

Figure 4. Temperature surface mode magnitudes for first three modes, mode 1 in purple/solid, mode 2 in green/dashed, mode 3 in orange/dotted for **(A)** estimate using three oldest cores and **(B)** estimate using all cores.

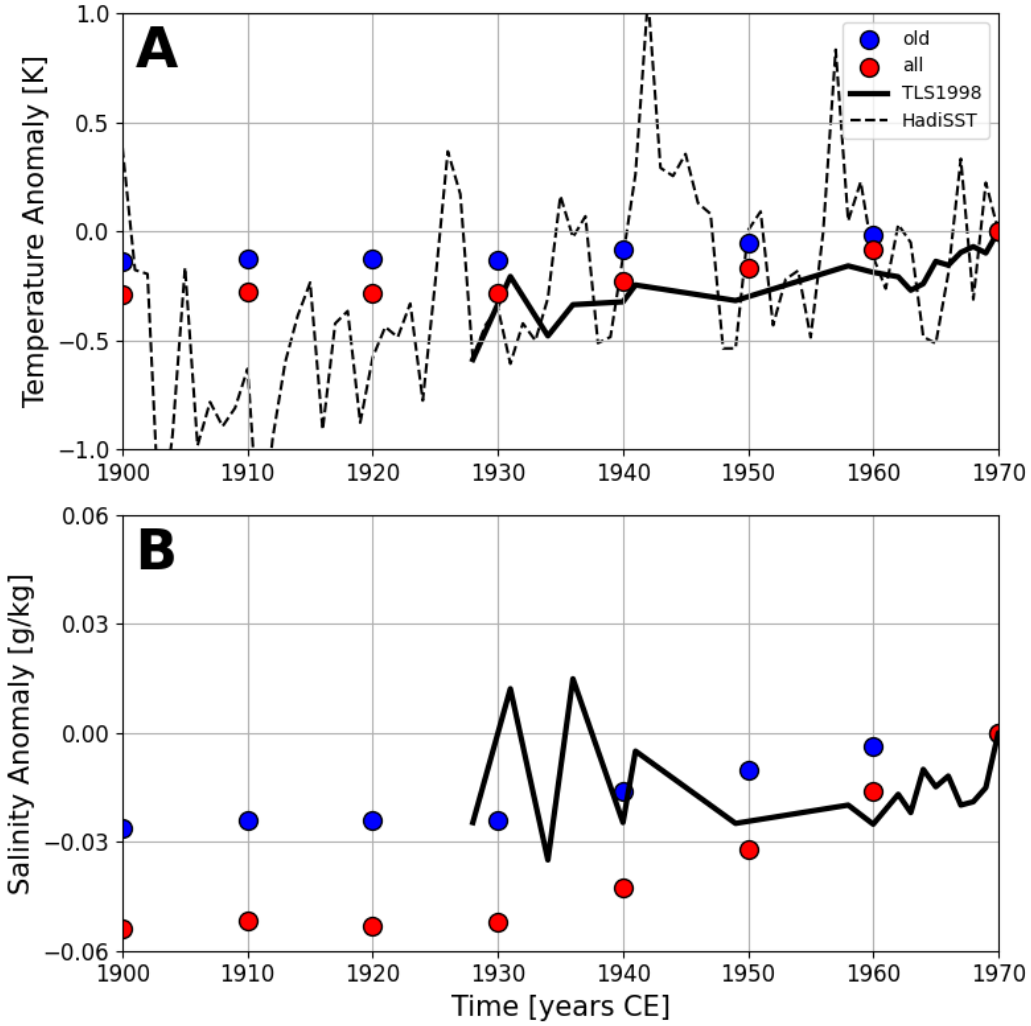

Figure 5. **(A)** Reconstructed Labrador Sea (60°W-30°E, 55°N-64°N) temperature and **(B)** salinity (computed as $0.5255 \times \delta^{18}O_{seawater}$) over the observational period (1850-1970) is shown for "old" (blue) and "all" (red) solutions. Solid line shows digitized Labrador Sea record from The Lab Sea Group, 1998 publication. Dashed line shows HadiSST Labrador Sea spatial mean (60°W-30°E, 55°N-64°N). All records are shown as an anomaly from 1970.

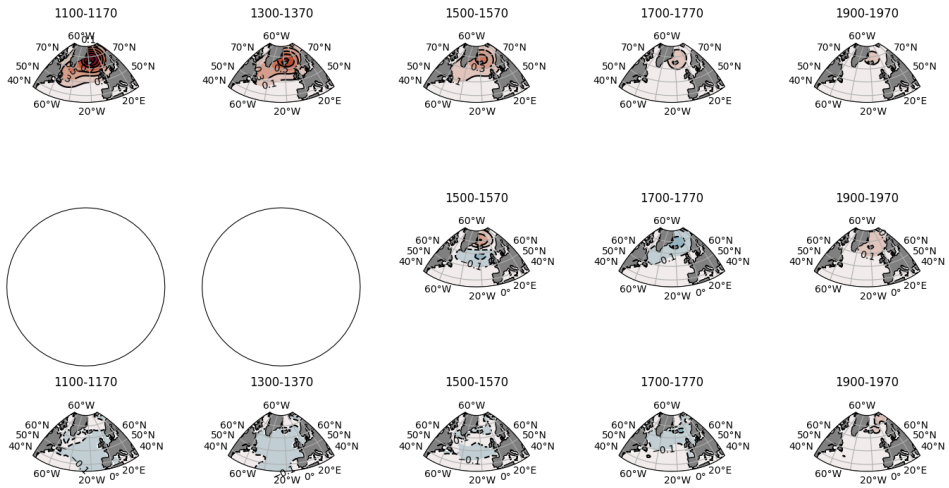

Figure 6. Reconstruction of surface potential temperature, 70-year mean, SVD approach, anomaly from 1850-1970. Top row is for old cores, middle row is for all cores, third row is Last Millenium Reanalysis SSTs (Hakim et al., 2016).

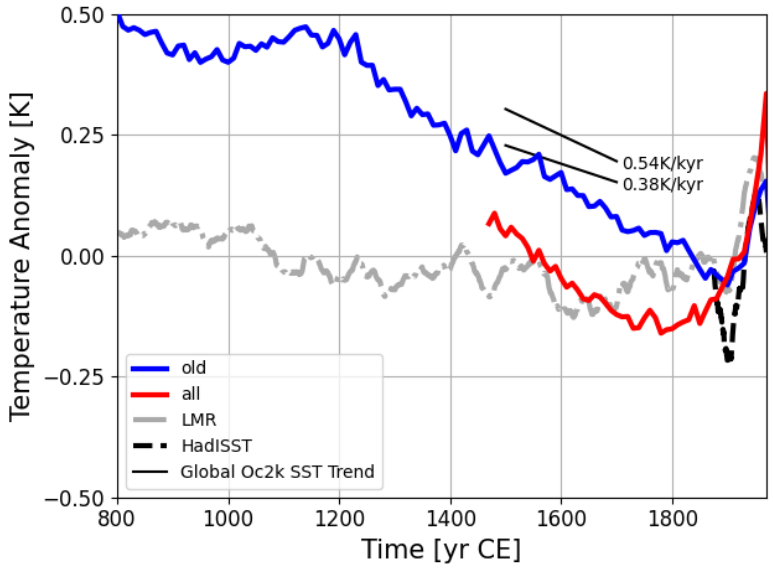

Figure 7. Mean North Atlantic temperature for "all" solution (red) and "old" (blue), Last Millenium Reanalysis (gray dash-dot), HadiSST (black dotted). Ocean2k trend of 0.54-0.38 °C/kyr (from Supplemental Figure 6) is shown in shaded gray region. All records are shown as anomaly from 1850-1970. For HadiSST and both reconstructions, the mean is computed over the region (50°W-20°E, 50°N-90°N)

[This will go in discussion, but for now: Estimated LIA N. Atl. cooling is calculated by computing the linear least squares slope between the maxima and minima occurring between 1000CE and 1900CE. Old: 0.67 ± 0.56 K/kyr, All: 0.77 ± 1.8 K/kyr]

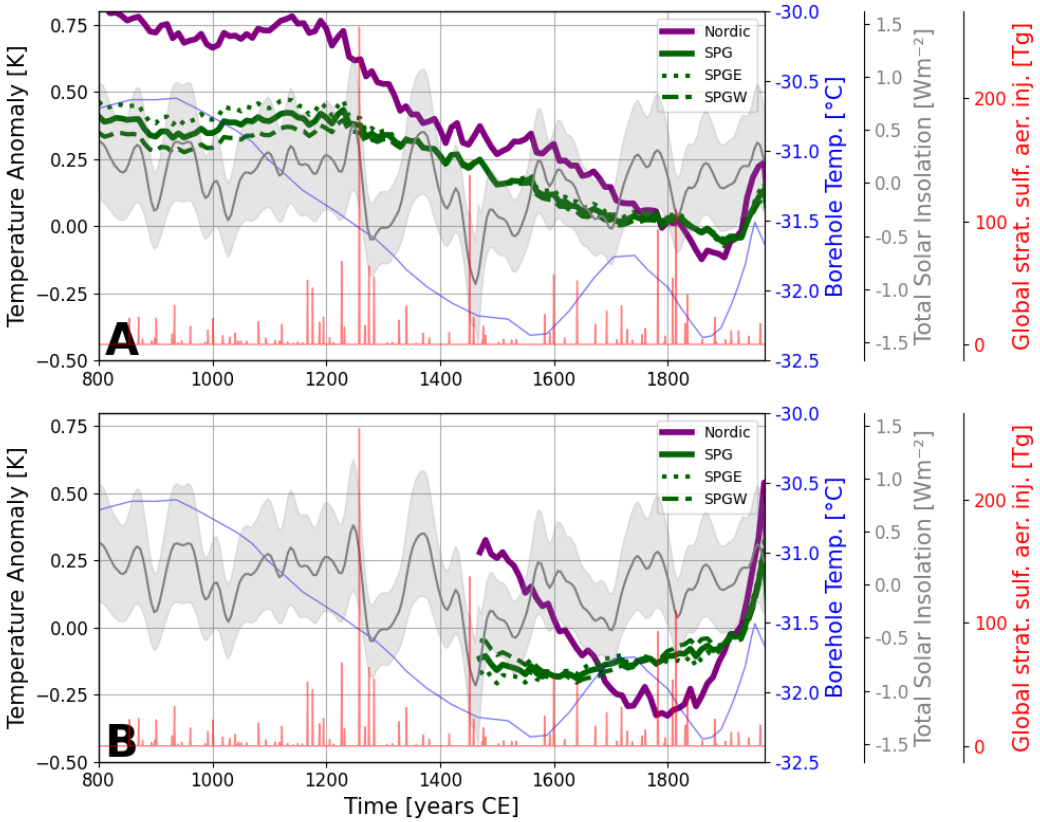

Figure 8. Temperature for **(A)** "old" and **(B)** "all" solutions observed for spatial averages in Nordic Seas (purple: 20°W-20°E, 65°N-80°N), the subpolar gyre (green: 60°W-0°, 50°N-65°N), the western subpolar gyre (green, dashed: 60°W-30°W, 50°N-65°N), and the eastern subpolar gyre (green, dotted: 30°W-0°, 50°N-65°N). All reconstructions are shown as anomaly from 1850-1970CE. Blue line shows digitized GRIP borehole record (Dahl-Jensen et al., 1998), grey line and shading show total solar insolation estimate with 1σ uncertainty (Steinhilber et al., 2009), and red lines show global stratospheric sulfate aerosol injection (Gao et al., 2008).
This figure is definitely too busy, I just thought it would be interesting to compare these records to some common forcing plot and see if anything pops out.
[For discussion: "old" solution reconstructs 0.9 ± 2.7K Nordic Sea cooling from 800-1860CE, and 0.5 ± 2.2K subpolar gyre cooling from 1230-1900CE.]

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